

PROBABILITY THEORY AND RANDOM PROCESSES (MA225)

LECTURE SLIDES

Lecture 12

Examples

Example 1: $E\left(\sum_{i=1}^n a_i X_i\right) = \sum_{i=1}^n a_i E(X_i)$ for real constants a_i .

Example 2: At a party N men throw their hats into the center of a room. The hats are mixed up and each man randomly selects one. Find the expected number of men who select their own hat.

Some Remarks

Remark: (X, Y) is discrete random vector iff X and Y are discrete random variables.

Remark: If (X, Y) is continuous random vector, then X and Y are continuous random variables.

Remark: If (X, Y) is continuous random vector, then

$$P((X, Y) \in A) = \int \int_{(x,y) \in A} f(x, y) dx dy,$$

for all $A \subseteq \mathbb{R}^2$ such that the integration is possible.

Some Remarks

Remark: (X, Y) may not be a continuous random vector even if X and Y are continuous random variables.

Remark: In general, if there exists a set A in $\mathbb{R}^2$ whose area is zero and $P((X, Y) \in A) > 0$, then (X, Y) does not have a JPDF.

Remark: If the joint distribution is known, then the marginal distributions can be recovered. However, the converse is not true.

Example 3: Let f and g be two PDFs and F and G be the corresponding CDFs. Define, for $-1 < \alpha < 1$,

$$h(x, y) = f(x)g(y) \{1 + \alpha(1 - 2F(x))(1 - 2G(y))\}.$$

Then h is a JPDF whose marginals are f and g .

Independent Random Variables

Def: The random variables $X_1, X_2, \dots, X_n$ are said to be independent if

$$F_{X_1, X_2, \dots, X_n}(x_1, x_2, \dots, x_n) = \prod_{i=1}^n F_{X_i}(x_i),$$

for all $(x_1, x_2, \dots, x_n) \in \mathbb{R}^n$.

Remark: X and Y are independent iff the events $E_x = \{X \leq x\}$ and $F_y = \{Y \leq y\}$ are independent for all $(x, y) \in \mathbb{R}^2$.

Remark: For DRV/CRV (X, Y) , the condition of independence is equivalent to

$$f_{X,Y}(x, y) = f_X(x)f_Y(y) \text{ for all } (x, y) \in \mathbb{R}^2.$$

Independent Random Variables

Theorem: If X and Y are independent, then

$$E(g(X)h(Y)) = E(g(X))E(h(Y)),$$

provided all the expectations exist.

Def: The covariance of two random variables X and Y is defined by

$$\text{Cov}(X, Y) = E[(X - E(X))(Y - E(Y))] = E(XY) - E(X)E(Y).$$

Def: The correlation coefficient of X and Y is defined by

$$\rho(X, Y) = \frac{\text{Cov}(X, Y)}{\sqrt{\text{Var}(X)\text{Var}(Y)}}.$$

Remark: If X and Y are independent, then $\text{Cov}(X, Y) = 0$. The converse is not true in general.

Remark: $|\rho(X, Y)| \leq 1$.

Remark: $\text{Cov}(X, X) = \text{Var}(X)$.

Remark: $\text{Cov}(X, Y) = \text{Cov}(Y, X)$.

Remark: $\text{Cov}(aX, Y) = a\text{Cov}(X, Y)$.

Remark: $\text{Cov}(X + Z, Y) = \text{Cov}(X, Y) + \text{Cov}(Z, Y)$.

Remark: $\text{Cov}\left(\sum_{i=1}^n a_i X_i, \sum_{j=1}^m b_j Y_j\right) = \sum_{i=1}^n \sum_{j=1}^m a_i b_j \text{Cov}(X_i, Y_j)$.

Remark: $\text{Var}\left(\sum_{i=1}^n X_i\right) = \sum_{i=1}^n \text{Var}(X_i) + 2 \sum_{i < j} \text{Cov}(X_i, Y_j)$.

Remark: If X_i 's are independent, then $\text{Var}\left(\sum_{i=1}^n X_i\right) = \sum_{i=1}^n \text{Var}(X_i)$.